Run Book

Deploying ADCS Certificate Expiry Scripts for Telegraf Agent Collection into Aria Operations

Microsoft AD CS / Windows CA local script deployment, validation, and Aria Operations Custom Script configuration

Document purpose	Deploy the supplied PowerShell metric and report scripts to a Windows Certificate Authority server and configure Aria Operations product-managed Telegraf agent Custom Script objects to collect certificate expiry metrics.
Target platform	Windows Server CA host running Microsoft AD CS, with Aria Operations Manage Telegraf Agents workflow already in use.
Metric scripts covered	Collection status, expiring certificate count (60 days), warning certificate count (30 days), critical certificate count (14 days), soonest days to expiry (60 days).
Companion report	Get-ADCSExpiringCertificateReport.ps1 for local operator review of the certificates that triggered warning or critical metrics.

Design note: These scripts are intended for the Aria Operations product-managed Custom Script workflow. Each metric script must return a single integer only. Detailed certificate information is deliberately separated into the companion report script.

1. Overview

This run book provides the deployment procedure for the ADCS PowerShell metric scripts and the supporting certificate detail report script. The approach assumes the scripts are copied to the local Windows Certificate Authority server and executed by the product-managed Telegraf agent Custom Script mechanism inside Aria Operations.

The metric scripts are intentionally lightweight and return a single integer each so that Aria Operations can ingest them reliably. The separate report script produces a human-readable HTML report and supporting files so operators can quickly identify which issued certificates are approaching expiry.

2. Script package and roles

File	Purpose	Expected output
ADCSMetricLib.ps1	Shared functions used by all metric scripts. Handles local certutil queries, revoked certificate exclusion, and date parsing.	Not executed directly
Get-ADCSCollectionStatus.ps1	Validates that the local CA query succeeds.	1 = success, 0 = failure
Get-ADCSExpiringCertificateCount60Days.ps1	Returns the number of issued, non-revoked certificates expiring in 60 days or less.	Count or -1 on failure
Get-ADCSWarningCertificateCount30Days.ps1	Returns the number of issued, non-revoked certificates expiring in 30 days or less.	Count or -1 on failure
Get-ADCSCriticalCertificateCount14Days.ps1	Returns the number of issued, non-revoked certificates expiring in 14 days or less.	Count or -1 on failure
Get-ADCSSoonestDaysToExpiry60Days.ps1	Returns the smallest days-to-expiry value found inside the 60-day window.	Days remaining, -1 if none, -2 on failure
Get-ADCSExpiringCertificateReport.ps1	Generates the operator-facing detailed certificate expiry report in the reports sub-folder.	HTML / CSV / TXT report files

3. Prerequisites

- The Windows server is a Microsoft AD CS Certificate Authority host and the scripts will run locally on that host. No -config parameter is used in this design.
- The Aria Operations product-managed Telegraf agent is already installed on the target Windows server and the server is visible in Manage Telegraf Agents.

- PowerShell script execution is permitted for the chosen deployment method. The recommended prefix in Aria is the full PowerShell path followed by -File.
- The operator has permissions to copy the PowerShell files to the target server and to create Custom Script objects in Aria Operations.
- A folder such as C:\Scripts exists or will be created to store the metric scripts, shared library, report script, log file, and generated reports.

4. Target folder structure

Create or confirm the following structure on the CA server:

```
C:\Scripts\
  ADCSMetricLib.ps1
  Get-ADCSCollectionStatus.ps1
  Get-ADCSExpiringCertificateCount60Days.ps1
  Get-ADCSWarningCertificateCount30Days.ps1
  Get-ADCSCriticalCertificateCount14Days.ps1
  Get-ADCSSoonestDaysToExpiry60Days.ps1
  Get-ADCSExpiringCertificateReport.ps1
  ADCSMetricErrors.log   (created automatically if required)
  reports\             (created by the report script)
```

5. Deployment procedure on the CA server

1. Copy all supplied PowerShell files into C:\Scripts so that the metric scripts and ADCSMetricLib.ps1 remain in the same folder.
2. Confirm the files are present by opening PowerShell and running Get-ChildItem C:\Scripts*.ps1.
3. If required by local policy, unblock the files after copy using Unblock-File against the script folder.
4. Do not rename ADCSMetricLib.ps1 unless you also update the dot-sourcing line in every metric script and in the report script.
5. Do not place the metric scripts in different folders from ADCSMetricLib.ps1; the scripts rely on $PSScriptRoot to load the shared library.

6. Local validation before Aria configuration

Run the following test block locally on the CA server. Each command should return a single integer only:

```
& {
  Write-Host "CollectionStatus:"
  powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\Scripts\Get-ADCSCollectionStatus.ps1"

  Write-Host "ExpiringCertificateCount60Days:"
  powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\Scripts\Get-
ADCSExpiringCertificateCount60Days.ps1"

  Write-Host "WarningCertificateCount30Days:"
  powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\Scripts\Get-
ADCSWarningCertificateCount30Days.ps1"

  Write-Host "CriticalCertificateCount14Days:"
  powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\Scripts\Get-
ADCSCriticalCertificateCount14Days.ps1"

  Write-Host "SoonestDaysToExpiry60Days:"
  powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\Scripts\Get-
```

```
ADCSSoonestDaysToExpiry60Days.ps1"
}
```

Metric	Validation guidance
CollectionStatus	Expected value is 1 when the local certutil query succeeds.
Expiring / Warning / Critical counts	Expected values are zero or greater when the query succeeds. Negative values indicate failure handling.
SoonestDaysToExpiry60Days	Zero or greater means a certificate was found in the 60-day window. -1 means none in window. -2 means failure.

7. Configure the Aria Operations Custom Script objects

Use the Manage Telegraf Agents workflow already established in your environment. Create one Custom Script object per metric script.

Recommended configuration values:

Display name	File path	Operational note
ADCS Collection Status	C:\Scripts\Get-ADCSCollectionStatus.ps1	Use to detect collection failures.
ADCS Expiring Certificate Count 60 Days	C:\Scripts\Get-ADCSExpiringCertificateCount60Days.ps1	Primary volume metric for all certificates in scope.
ADCS Warning Certificate Count 30 Days	C:\Scripts\Get-ADCSWarningCertificateCount30Days.ps1	Warning threshold metric.
ADCS Critical Certificate Count 14 Days	C:\Scripts\Get-ADCSCriticalCertificateCount14Days.ps1	Critical threshold metric.
ADCS Soonest Days To Expiry 60 Days	C:\Scripts\Get-ADCSSoonestDaysToExpiry60Days.ps1	Most useful countdown metric for urgency.

For each Custom Script object, use the following base settings:

- Prefix: %SYSTEMROOT%\System32\WindowsPowerShell\v1.0\powershell.exe -File
- Args: leave blank
- Timeout: 5 minutes is typically suitable
- Status: Activated

8. Verify metric collection in Aria Operations

- After saving each Custom Script object, wait for at least one collection cycle and then refresh the object or metric view.
- Confirm the metric is no longer in an old-data state and is receiving fresh values.
- Verify that the values shown in Aria match the results from the local PowerShell validation on the server.
- If the metric object appears but values do not update, review the script file path, PowerShell prefix, and local script execution under the same account context used by the agent.

9. Recommended alert mapping

Metric	Suggested condition	Operational intent
ADCS Collection Status	< 1	Raise a collection failure alert.
ADCS Critical Certificate Count 14 Days	> 0	Raise a critical certificate expiry alert.
ADCS Soonest Days To Expiry 60 Days	>= 0 and <= 14	Raise or support a critical urgency alert.
ADCS Warning Certificate Count 30 Days	> 0	Raise a warning certificate expiry alert.
ADCS Soonest Days To Expiry 60 Days	> 14 and <= 30	Raise or support a warning urgency alert.

10. Deploy and use the detailed certificate report

The report script does not feed a metric directly into Aria. Instead, it gives an operator a local report that explains which certificates are behind warning or critical alerts.

- Run the report script manually on the CA server when an alert is raised or during routine review.
- The script writes its output into the reports sub-folder beneath the script location.
- The HTML output provides the operator-facing view, while the CSV export allows additional filtering or archive handling.

Example command:

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\Scripts\Get-ADCSExpiringCertificateReport.ps1"
```

11. Troubleshooting

Check	Action
Metric returns fallback value	Review C:\Scripts\ADCSMetricErrors.log and rerun the local validation block.
No current data in Aria	Check the Custom Script prefix, file path, timeout, and local PowerShell execution under the agent context.
Script cannot find shared library	Confirm ADCSMetricLib.ps1 is in the same folder as the metric and report scripts.
Report generation fails	Run the report script manually in PowerShell and confirm the reports sub-folder can be created.
Unexpected certificate count	Manually test certutil output on the CA and confirm revoked certificates are being excluded as intended.

Appendix A. Quick operator commands

Command purpose	Command
List script files	`Get-ChildItem C:\Scripts\*.ps1`
Run the 60-day report	`powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\Scripts\Get-ADCSExpiringCertificateReport.ps1"`
Read the shared error log	`Get-Content "C:\Scripts\ADCSMetricErrors.log" -Tail 50`